

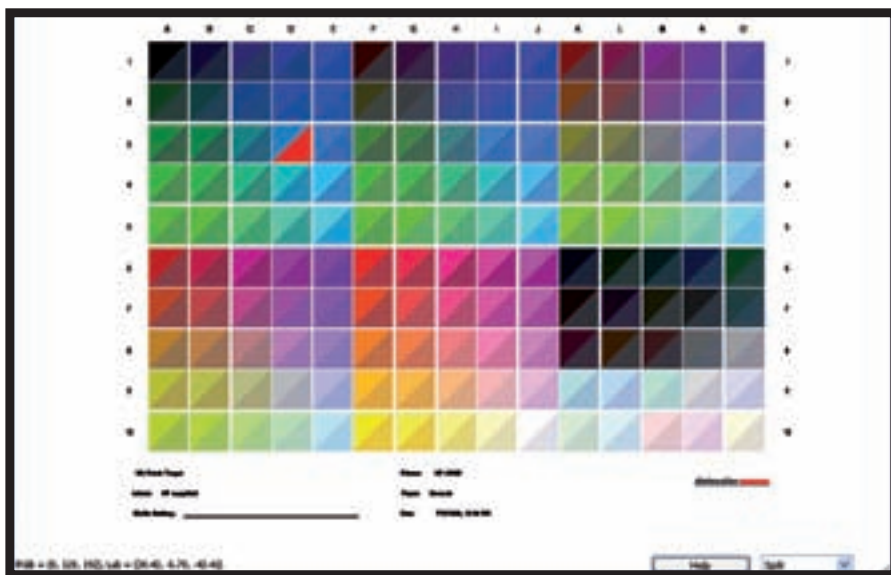
Cyan-Magenta-Yellow

This is the subtractive colour system used by printers: cyan is white minus all the red frequencies, magenta is white minus all the green yellow is white minus all the blue

Is vision 23-bit?

It is considered that human vision can distinguish up to ten million colours – a little more than 2^{23}

Feature



The “Build Profile Setup” adjustment screen – a clear indication that printer colour management works imperfectly and still requires subjective adjustment.

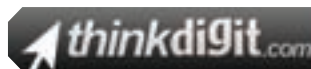
you place the printed test sheet on a table, and on top of this the guide – this helps you position the Spyder sensor over each patch. You start in the top left corner, and progress along each row. A cursor in the shape of a red triangle on screen shows which patch is being measured, and once the measurement has been taken – either by pressing the button on the Spyder or by hitting Enter – the software makes a sound to indicate the measurement, and then another as it automatically moves the cursor along to the next patch.

The user has to take care to move the sensor itself, and make sure it is correctly positioned over each patch. It is easy to keep to the correct patch in the sequence, but care must be taken to ensure the positioning is correct and that the sensor is flat on the paper – otherwise bad measurements will result. It took just over ten minutes to measure all 225 patches in the Fast target – it is well worth doing this slowly and very carefully, although you can go back and retake measurements,

if needed. We tried this a few times, and found that measurements were always varying slightly.

The next screen (called “Build Profile

Setup”) gives a set of further options for adjustment before the actual profile is built. We’ll come back to this. You enter a name for the file, select any adjustments as necessary, and then press “Next”. The profile is built and saved to disk. You then come to a screen named “SpyderProof”. This allows you to print and compare the results. You can here choose the rendering intent, print the sample, and compare with the image on screen, assuming you have a calibrated screen. It would be a much better idea for actual sample printouts to be provided by Datacolor here – these would enable a much more accurate comparison.



Discuss this article at
www.thinkdigit.com/d/998971

WHY BOTHER WITH COLOUR MANAGEMENT?

In an ideal world, if you take a photograph of any scene (or a scan of any image), the colours you see when directly looking at the scene would be indistinguishable from those that you see when that image is displayed on your monitor or printed on your printer. This is clearly critical for high-end applications such as book and magazine publication, but can also be important for semi-professional photography or enthusiastic amateurs.

But it is a far from perfect world, and there is no such thing as a perfect camera, screen, printer, etc.

Not only do the colours not match exactly, but also the range of colours – the gamut – that any device can detect or reproduce will vary from one device to another; and all devices have gamuts that are considerably less than that perceived by human vision.

The process of making adjustments as colours are mapped from one device to another so as to bring the colours displayed or printed as close as possible to those perceived in the original scene is called colour management.

In our tests, we used a standard PDI test file, which certainly seemed to give better results when printed using the created profile than not. However, we also printed some images of artwork, and the results seemed best – most true to the original – when simply using the printer’s built-in Adobe RGB profile. Clearly, this is far from an exact science.

This type of variability is obviously the reason for the existence of the “Build Profile Setup” screen that immediately precedes the creation of a profile. It obviously makes sense the first time you profile your printer to make no other manual adjustments – let the software do its job first, as objectively as possible, and then only later make further manual adjustments.

I asked the distributors of the Spyder why these extra controls are there, and they came back with a mixture of answers, some of which involved personal preferences. The other main points concerned the physical characteristics of the printer and media themselves; these were, in brief:

The presence of optical brighteners in the paper can at times fool the machine and lead to yellowish greys. Matt fine art papers require compensation for shadow detail. Some printers (especially some lasers) are weak in printing highlight details. Profiling tends to work best in printers which have extra grey inks, in other printers you might get some colour cast which can be removed manually.

Conclusions

Profiling a display with the Spyder Elite is a simple process. After some initial setup and checks, you place the Spyder against the screen, and the software runs through various colour patches, taking measurements as it goes. The profile is then created and used

by the system. It is a good idea to repeat the calibration every few weeks, but that is basically it, certainly with a modern, good quality display.

Our experience certainly suggests that profiling your printer with a system such as the Datacolor Spyder is worth doing, but much more is needed in terms of user judgement and time. Straightforward objective profiling simply does not work properly. You need to become familiar with your printer, understand the types of printing that you need to do, and how the printer responds to different profile adjustments and media. It is best to treat a tool like the Spyder as an aid in understanding how to adjust your printer profile. If you think that you can run just one calibration and get excellent results, think again, unless, of course, you have stumbled upon that elusive near-perfect, printer-paper-ink combination. 

Getting Spyder

Contact: Eesh Dewan
Phone: +91-11-4153 9116
Email: info@bhushanphoto.com
Web site: www.bhushanphoto.com